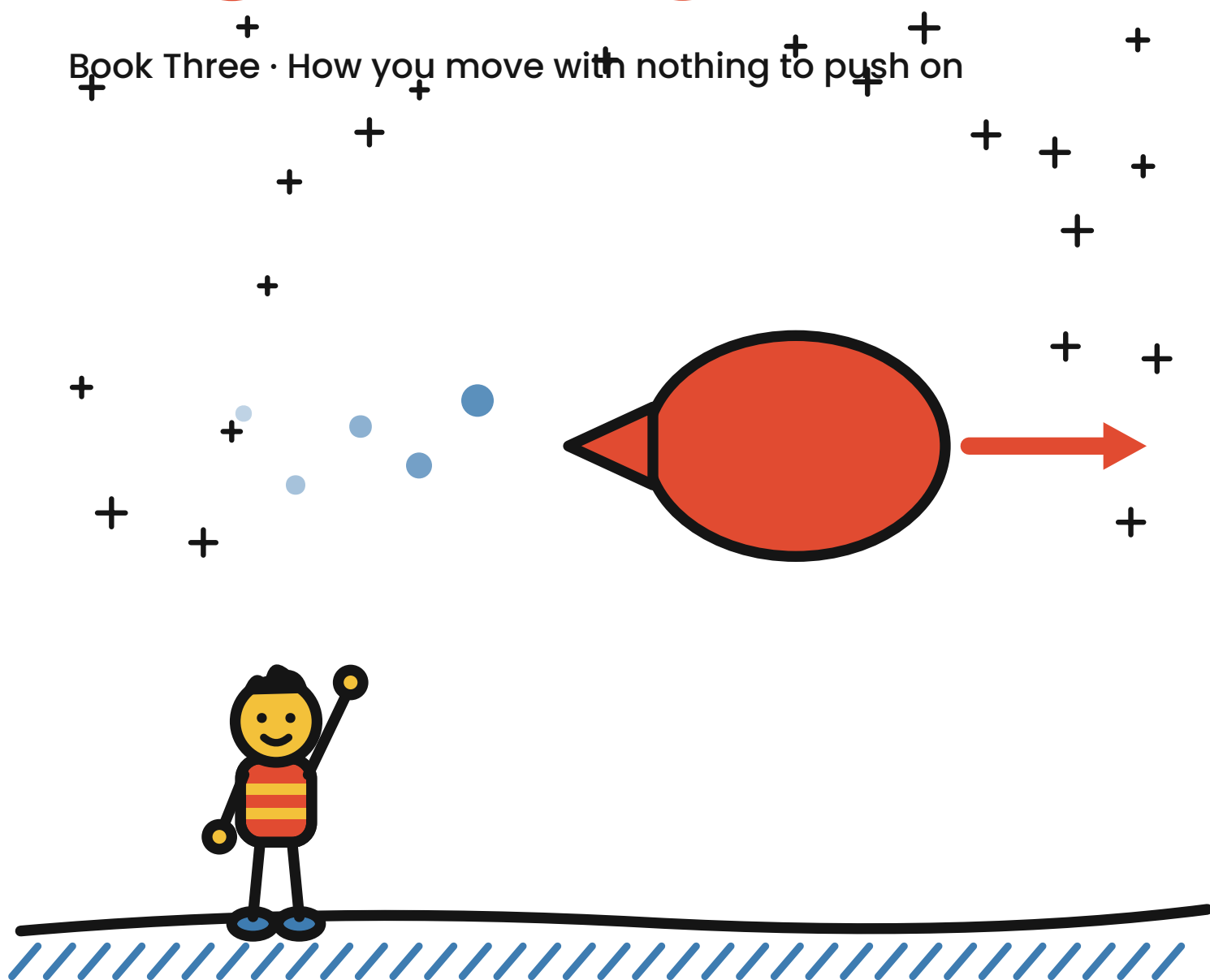


PUSH AGAINST NOTHING

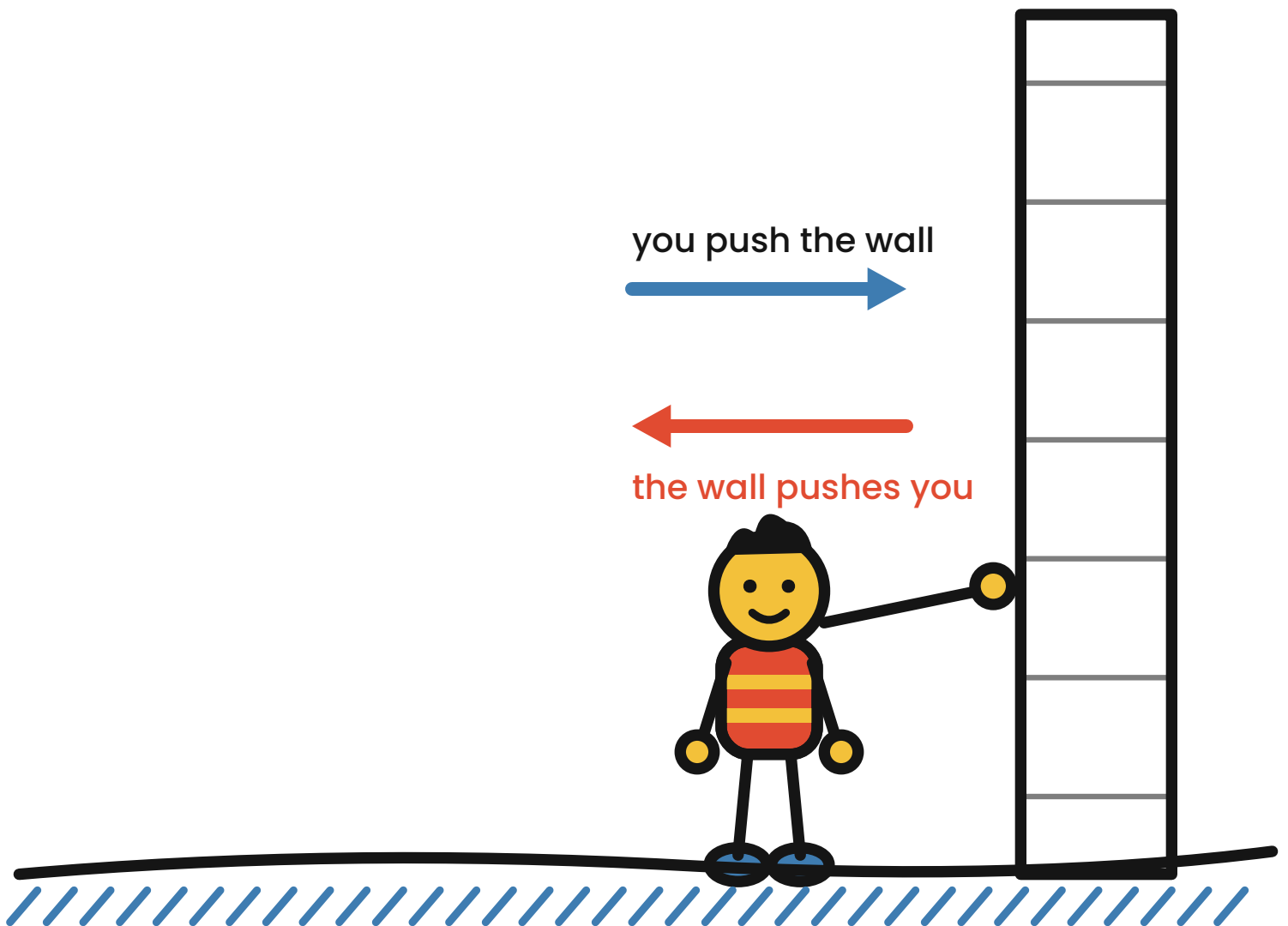
Book Three · How you move with nothing to push on



Print it. Read it together. Then find a balloon.

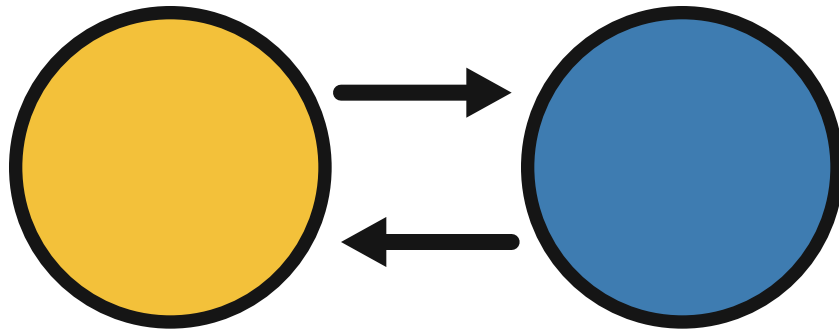
Push on a wall.

Go on. Push hard.



You can feel it pushing back.

Everything you push
pushes you back.

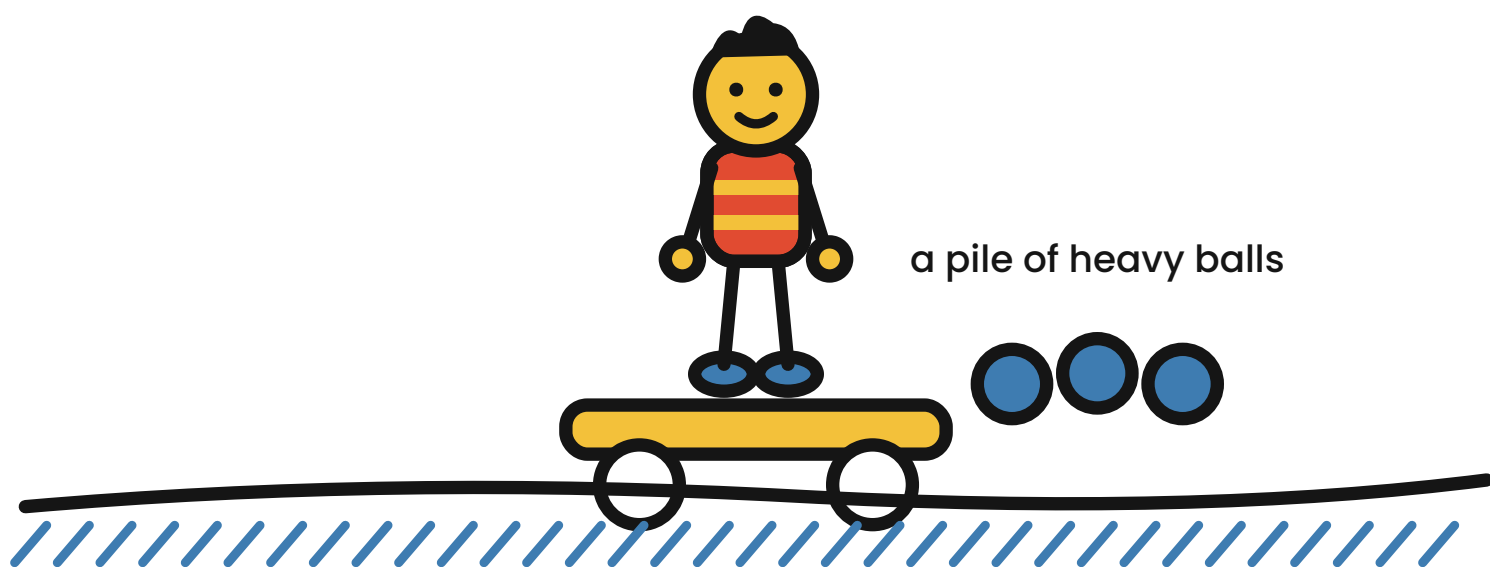


always both ways, always the same push



This is the rule the whole book is built on.

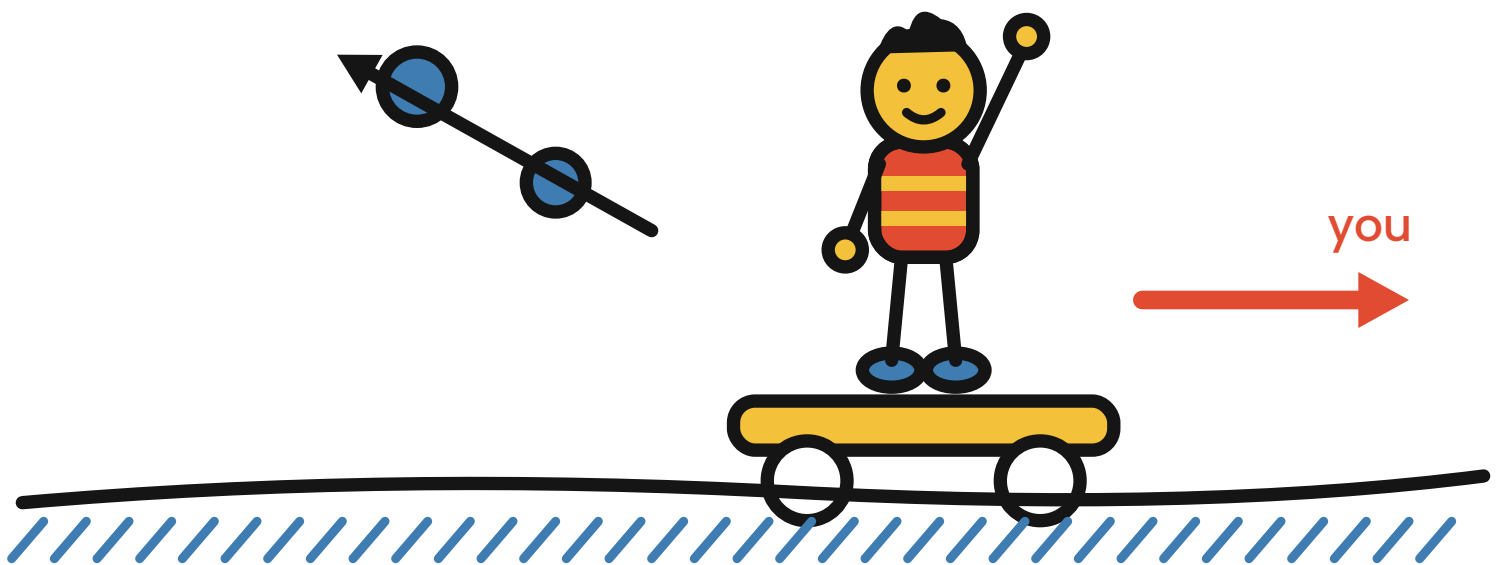
Stand on a board with wheels.



Nothing to hold. Nothing to push on.

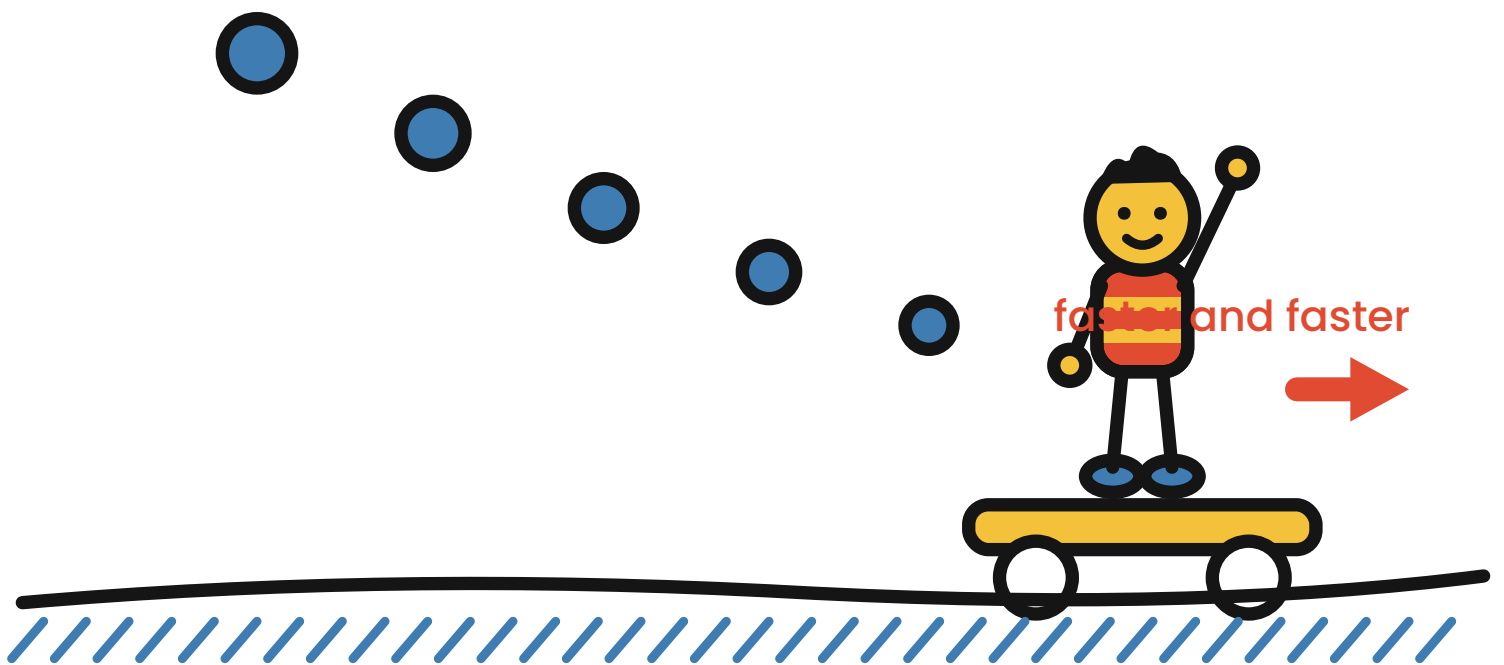
Throw one backwards.

What happens to you?



The ball goes that way. You go this way.

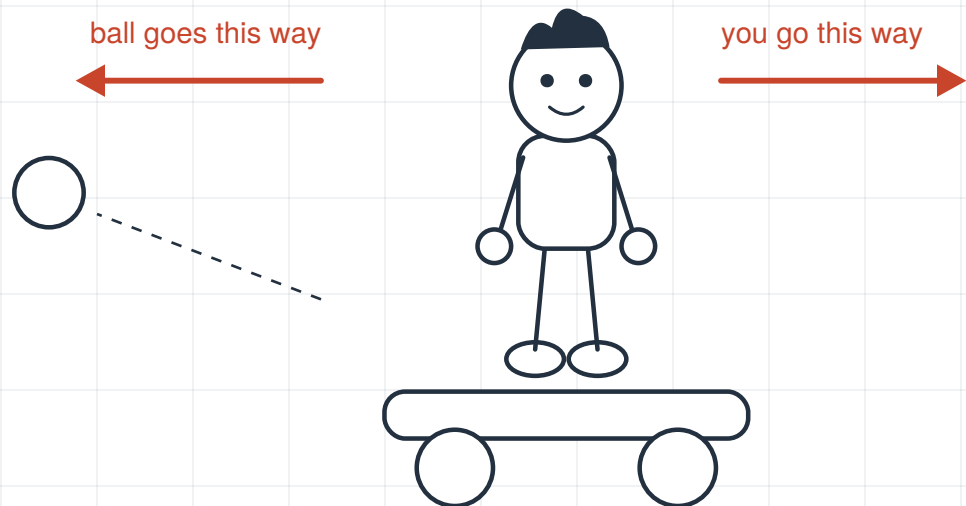
Throw another.
And another.



Every single throw makes you quicker.

Where does the speed come from?

Push Against Nothing · figure 1 · action and reaction

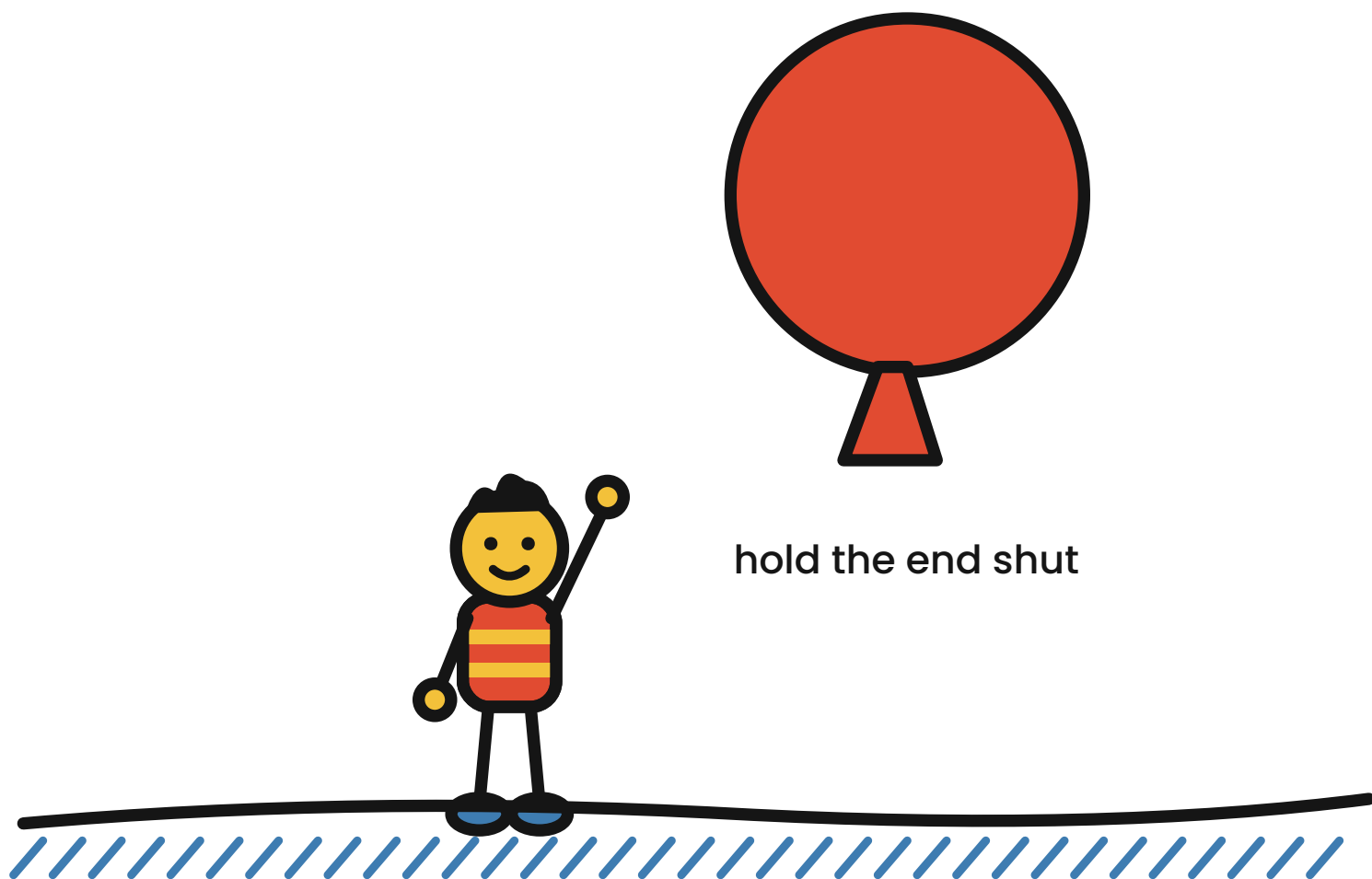


nothing is touching the ground —
no push from outside

A heavy ball thrown slowly, or a light ball thrown fast:
both move you the same amount. Only mass times speed counts.

A rocket takes the second option, and takes it to an extreme.

**Now blow up
a balloon.**

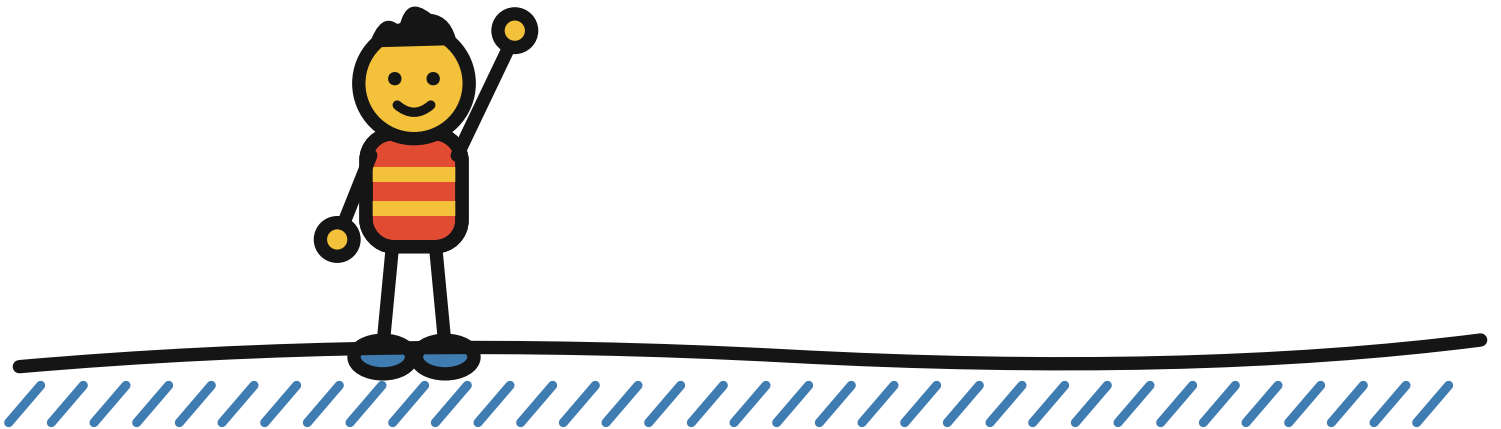
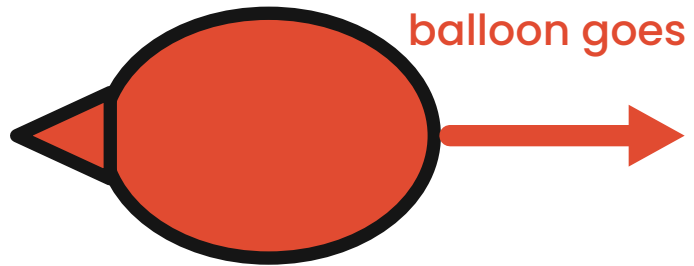


All that air is squeezed up inside.

Let it go.

air rushes out

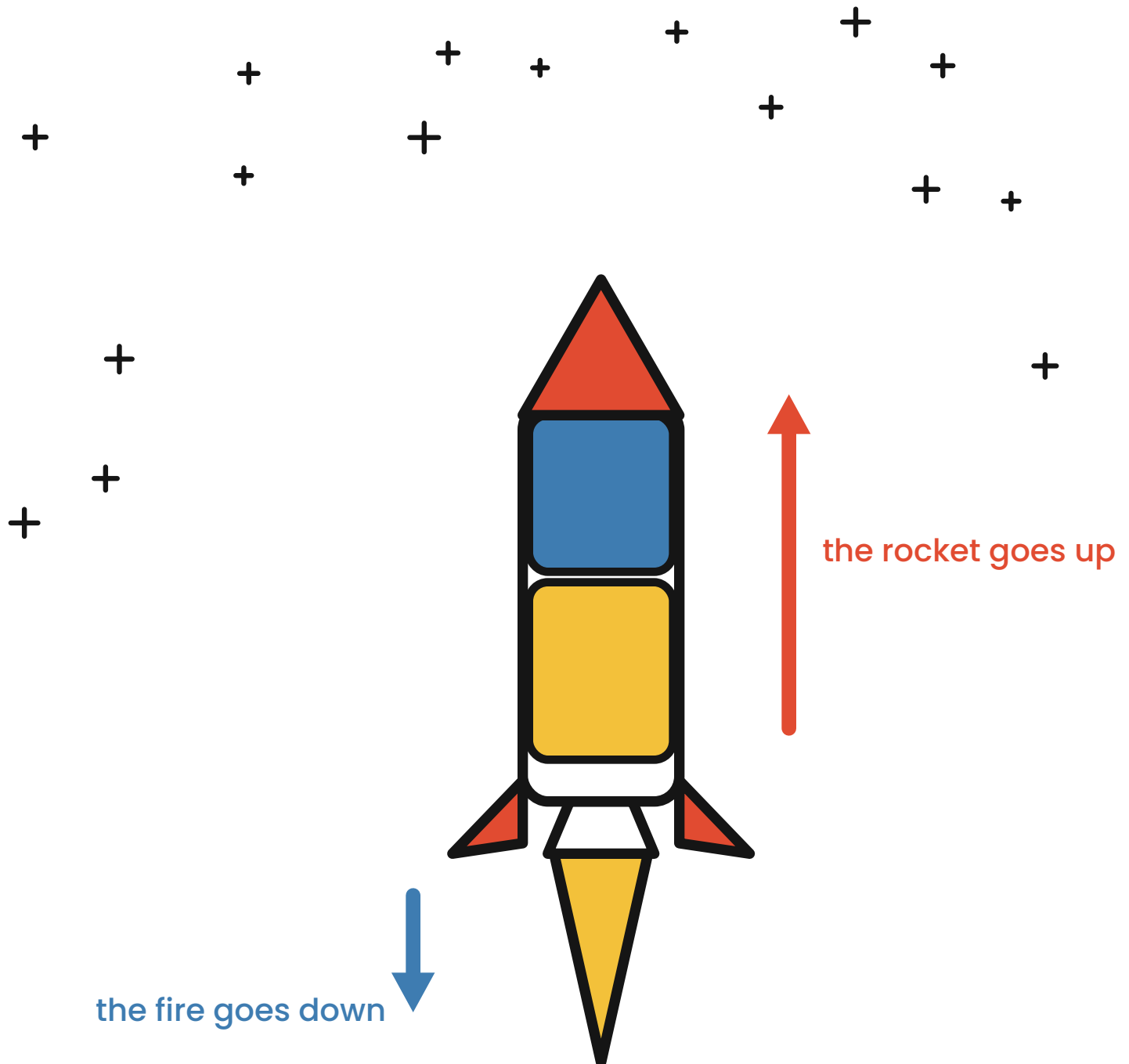
balloon goes



The balloon pushes the air out.

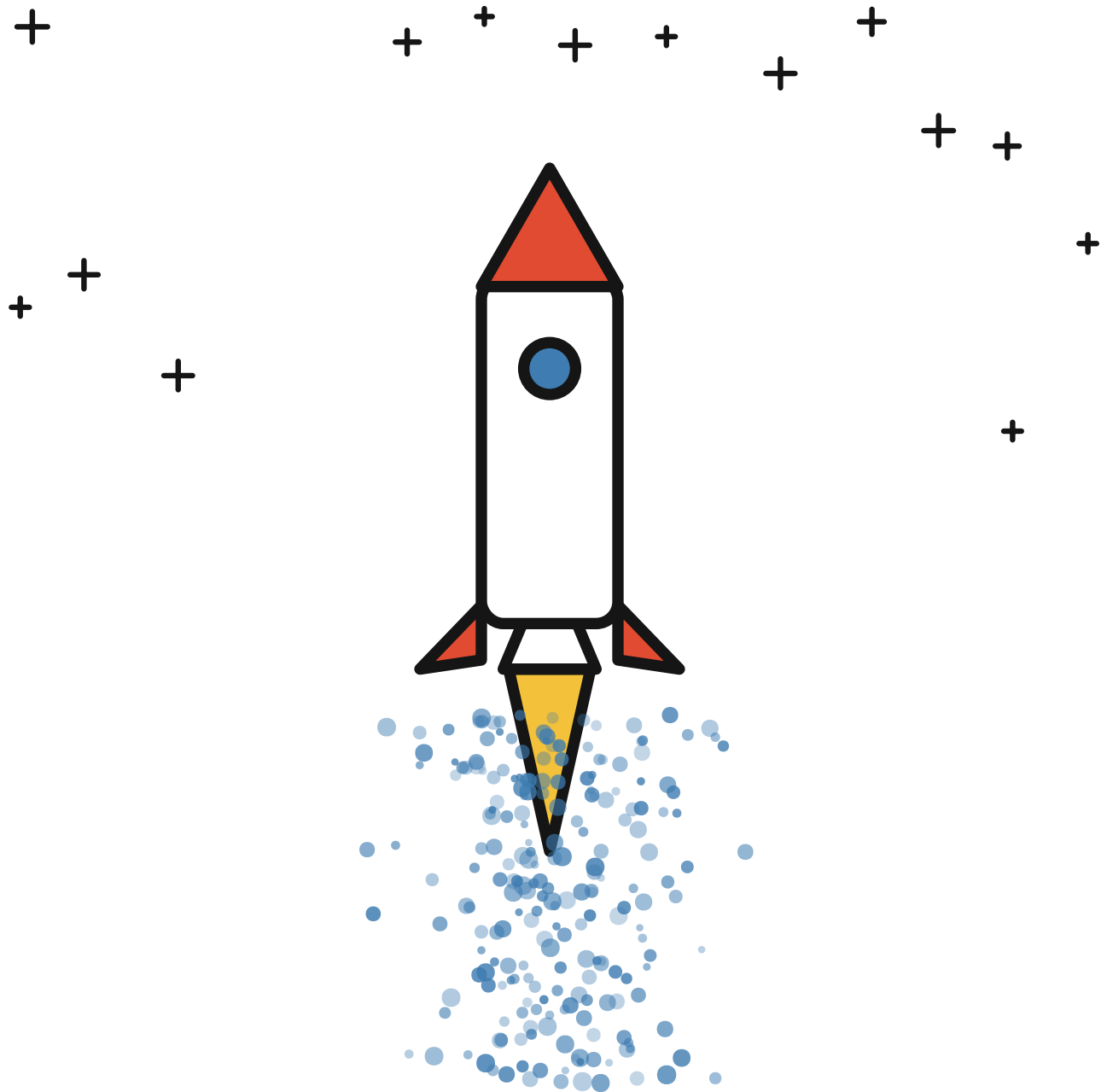
So the air pushes the balloon along.

A rocket is a balloon
that is on fire.



It throws hot gas down, so it goes up.

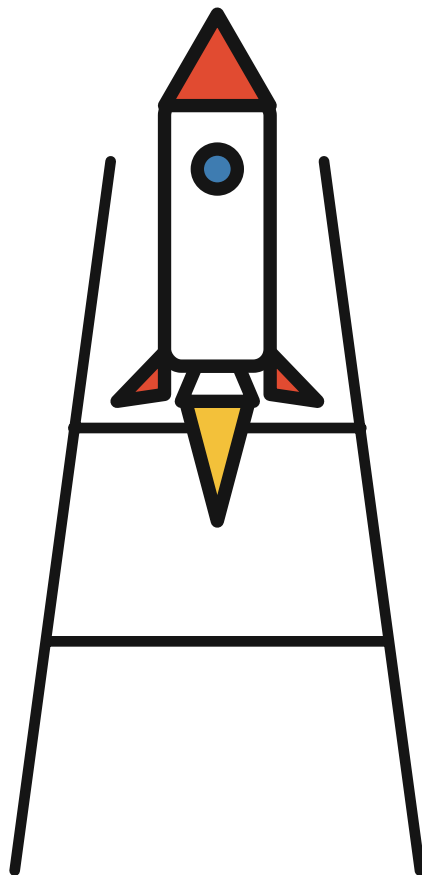
Not one throw.
Millions every second.



Every tiny bit of gas gives it one tiny push.

The first one was small.

A man called Robert Goddard built it.

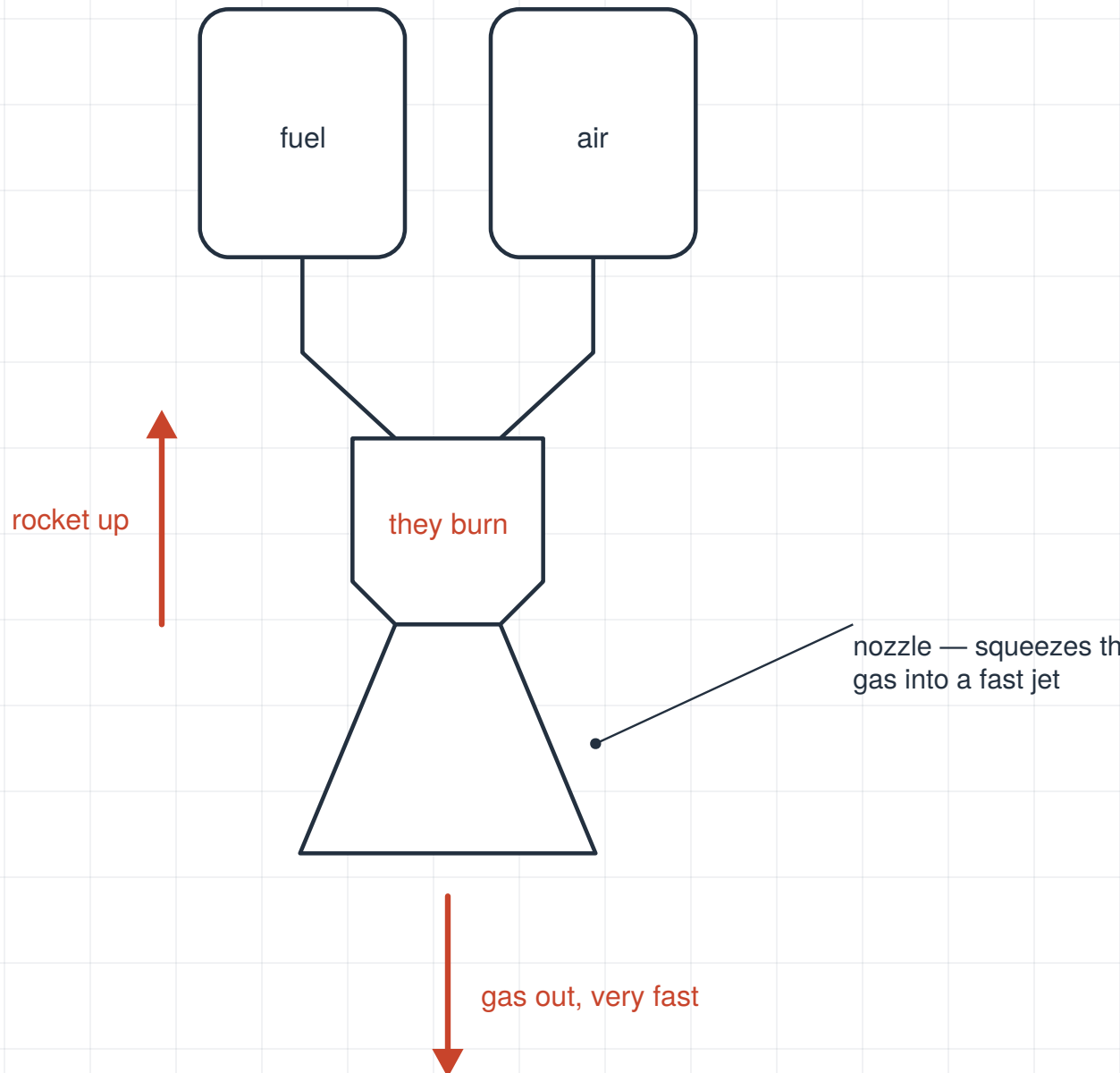


It flew for two and a half seconds,

and landed in a cabbage field.

Inside the engine.

Push Against Nothing · figure 2 · where the push is made



Thrust is simply how much mass leaves, times how fast it leaves.

Nothing outside the rocket is involved at any point.

Try this tonight.

1 Balloon on a string.

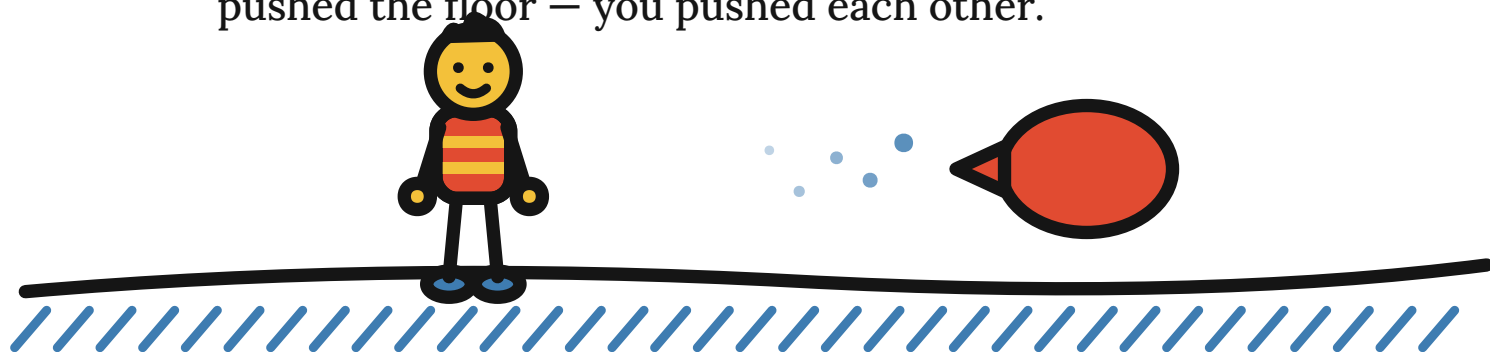
Thread a long string through a drinking straw and tie it across the room. Tape an inflated balloon to the straw, let go, and watch it fly the whole length. That is a guided rocket test.

2 Sit on a rolling chair.

Feet off the floor, then throw something heavy — a cushion, a shoe. You will roll backwards. Throw it harder and you roll faster. That is the whole of page five.

3 Push him, gently.

Stand on a smooth floor in socks and push each other's hands. You both slide apart. Neither of you pushed the floor — you pushed each other.



For the grown-up.

The one idea

Newton's third law is the entire engine principle. A rocket does not push on the air or on the launch pad. It throws mass overboard very fast, and the reaction accelerates what is left. Thrust equals mass flow rate times exhaust velocity — that is the whole equation.

Robert Goddard

He flew the first liquid-fuel rocket on 16 March 1926 at his aunt's farm in Auburn, Massachusetts. It burned for about two and a half seconds, rose roughly 12 metres, and came down in a cabbage patch 56 metres away. Every orbital rocket since is a descendant of that flight.

The newspaper story

In 1920 the New York Times mocked Goddard in an editorial, claiming a rocket could not work in a vacuum because it would have nothing to push against. They printed a correction in July 1969, the week Apollo 11 launched. Vacuum is in fact where rockets work best.

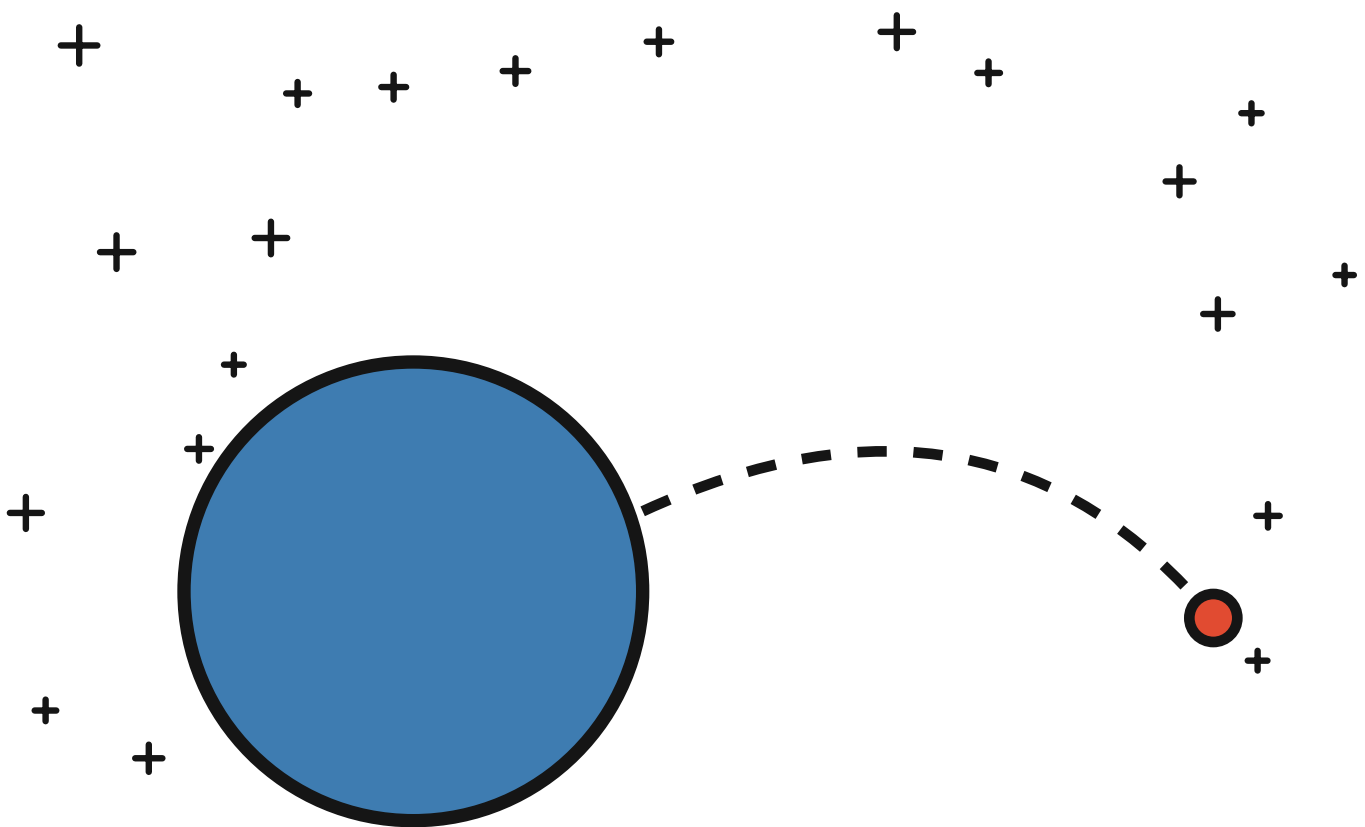
Why the balloon is the right demonstration

It is not an analogy. A balloon is a cold-gas rocket, the same class of device used for attitude control on real spacecraft. He is not being shown something like a rocket; he is being shown a rocket.

Next:

Falling and Missing

Book Four · why going up is only the easy half



Throw a ball hard enough
and it never comes down.